Practice Quiz One Solution: KCL, KVL, and Ohm's Law (Intermediate)

For the circuit shown below, with an unknown circuit element, find the following:

a) v_x

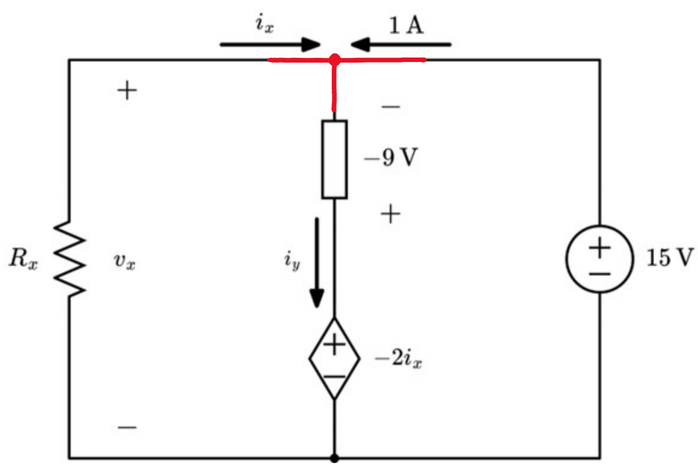
b) i_x

c) R_x

d) i_y

Finally, calculate the power of the unknown circuit element and state whether the power is absorbed or delivered.

Step One: KVL, right loop, following 1A:



$$-(-9)+(-2i_x)-15=0$$

$$-2i_x=6$$

$$i_x=-3\text{A}$$

Step Two: KVL, left loop, following i_x:

$$-(-9)+(-2i_x)-V_x=0 \qquad i_x=-3\text{A}$$

$$V_x=15\text{ V}$$

Also, you can identify that the Independent Voltage Source is in parallel with V_x

Step Three: Ohm's Law, observing passive sign convention:

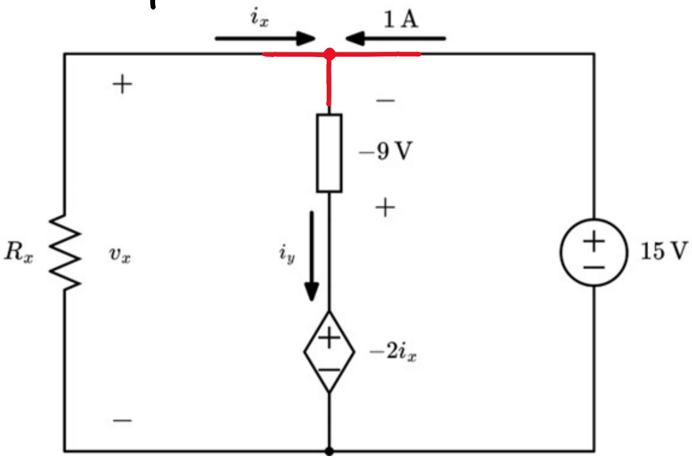
$$R_x=-\frac{V_x}{i_x}=-\frac{15\text{ V}}{-3\text{ A}}=5\Omega$$

Step Four: KCL, highlighted node:

$$-i_x-1+i_y=0$$

$$i_y=-2\text{A}$$

Step Five: Power Calculation



$$P_{element} = (-9V)\cdot(-(-2A))$$

$$P_{element} = -18\,W$$ Delivering

Self-Grading Rubric	
Correct KVL equation for the right loop	+1.5 points
Correct value for i_x	+1.5 point
Identify v_x = 15 V, either using a KVL equation or by inspection	+1.5 point
Correct value for i_y using a KCL equation	+1.5 points
Identify R_x = 5 Ω by using Ohm's law and Passive Sign Convention (no credit for answering R_x = -5 Ω)	+1.5 point
Correct value for the power of the unknown element ($P_{element}$)	+1.5 point
Correct sign for the power of the unknown element	+0.5 point
Mention that the unknown element is delivering power	+0.5 point